



Western Gorilla Vocal Repertoire and Contextual Use of Vocalizations

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Abstract

Despite considerable interest in the vocal communication of non-human primates and its possible relevance to theories of language evolution, we know surprisingly little about how vocal communication varies between closely related species inhabiting differing environments. Here, we provide the first quantitative description of the vocal repertoire, calling rates, and call usage in wild western gorillas and compare it to the previous work on mountain gorilla vocal behavior. During 1572 h of focal follows ($n = 533$), we collected behavioral data on and recorded vocalizations ($n = 2163$) of eight individuals in one group at Mondika Research Center (Republic of Congo). We supplemented these data with opportunistic recordings of an additional adult male in a second group. We used discriminant function analysis to test how well calls can be categorized by their acoustic structure and used behavioral data to determine the typical usage of western gorilla call types. The vocal repertoire comprised of 17 call types. Twelve of 17 call types were given primarily in a single context. Our results were similar to previous studies of mountain gorillas in that grunts, and grumbles were used most frequently and the silverback male vocalized more frequently than other group members. However, compared to mountain gorillas, western gorillas used an additional call type (sex-whinny), used a second call type (hoot series) in a completely different context and by all age–sex classes, and used many more call types in a more context-specific fashion. Our study suggests that although vocal production is highly constrained by morphology and phylogeny, differing social and ecological conditions can yield differences in the use and function of calls, even between two closely related species such as western and mountain gorillas.

Introduction

Communication is what separates or brings and maintains individuals together, creating and regulating social relationships that are critical for fitness and survival (Lucas & Freeberg 2010). While communication is multimodal, vocal signals are paramount for species such as non-human primates inhabiting forested environments, where visual cues are limited (e.g., Altmann 1967; Robinson 1981; Boinski & Campbell 1996; Uster & Zuberbühler 2001). The study

of the complete vocal repertoire, including comprehensive analysis of the behavioral context of vocalizations, is one way to investigate a species' vocal communication. Additionally, it allows comparison across species of call structure, meaning, and function (e.g., Morton 1977; Mitani 1996; Owings & Morton 1998; Owren & Bachorowski 2003) as well as of repertoire complexity and flexibility (McCowan et al. 2002; Fedurek & Slocombe 2011; Gustison et al. 2012), contributing to our understanding of the evolution of primate vocal systems and ultimately human language.

Early studies on great apes described and classified the vocal signals of chimpanzees (Reynolds & Reynolds 1965; van Lawick-Goodall 1968; Marler 1969), bonobos (de Waal 1988; Bermejo & Omedes 1999) and mountain gorillas (Schaller 1963) as part of their natural history. Interspecific comparisons of the vocal repertoire of these African apes documented that they produce acoustically similar calls, which may be broadly categorized into gross acoustic groups, including grunts, hoots, screams, and barks (Marler 1969; Mitani 1996). However, species differ in call rate and function, possibly as a result of differences in group cohesion and social structure across species (Marler 1976; Marler & Tenaza 1977; Mitani 1996). For instance, mountain gorillas live in cohesive groups and vocalize frequently (in average eight times per h), mostly using short-range grunts (Harcourt et al. 1993). In comparison, chimpanzees exhibit fission–fusion social dynamics and vocalize half as frequently (in average four times per h), mostly using long-range pant hoots (Clark & Wrangham 1993; Mitani 1996). Intraspecific comparison of vocal repertoires has also documented age–sex class and individual differences in both repertoire and vocal behavior (Fossey 1972; Harcourt et al. 1993; Mitani 1993; Hohmann & Fruth 1994). In both chimpanzees and gorillas, males are the highest-ranking individuals and have been reported to produce the largest vocal repertoire and to call more often than any other individual (reviewed in: Mitani 1996). In contrast, because females and infants are more vulnerable to predators and infanticide, they tend to call less frequently, using more fearful signals to seek protection and allies (Fossey 1972). In addition, as in other primates, apes produce calls in infancy (i.e., weaning calls) that they do not produce as adults (Fossey 1972; Goodall 1986).

In spite of the above studies, and considerable interest in the vocal communication of non-human primates and its relevance to theories of language evolution (Pinker & Bloom 1990; Zuberbühler 2003; Cheney & Seyfarth 2005; Seyfarth et al. 2010), we still know surprisingly little about vocal communication in great apes, humans' closest relatives (Slocombe et al. 2011). Among African apes, western gorillas (*Gorilla gorilla*) have remained largely unstudied and, as a consequence, their vocal repertoire is still unknown. Given the morphological and phylogenetic constraints inherent in vocal production (reviewed in: Hammerschmidt & Fischer 2008; Seyfarth et al. 2010), one might predict that the vocal repertoire of western gorillas is very similar to that of the closely related mountain gorilla and expect to identify the same 16 major vocalizations described for mountain gorillas

(Schaller 1963; Fossey 1972; Harcourt et al. 1993; summarized in Table 1). However, clear differences in ecological and social factors between the two gorilla species might affect their communication systems and consequently their vocal behavior (Snowdon & Hodun 1981; Waser & Brown 1986; Oda 1996; Sugiura et al. 2006; Ey et al. 2009). First, unlike mountain gorillas, which live in montane forests and travel short distances to forage on abundant herbs (Schaller 1963; Vedder 1984; Watts 2000; but see Rothman et al. 2007), western gorillas inhabit dense tropical lowland forests and travel greater distances to include much more fruit in their diet (Goldsmith 1999; Doran-Sheehy et al. 2004). Thus, due to the presumed reduction in visual cues in tropical forests (Morton 1975; Robinson 1981; Uster & Zuberbühler 2001; Zuberbühler 2006) compared to montane forests, one might expect that context-specific calls might be more common in western than mountain gorillas to facilitate within-group communication and coordination among individuals (e.g., Boinski 1991, 1993). As it has been proposed that signals that are strictly associated to particular contexts have evolved in circumstances with strong selective pressure (Hauser 1996), one might also suggest that western gorillas might use them prevalently in feeding, mating, agonistic (within group and between group) and potentially threatening contexts (Hauser 1996). Second, group spread has been suggested to be greater in western than mountain gorilla groups, possibly as a result of the aforementioned differences in diet and daily range (Tutin 1996; Doran & McNeillage 2001; Bermejo 2004). For instance, in mountain gorilla groups, all individuals are usually found within a 50 m radius of one another (Vedder 1984; Watts 1991b), while in western gorillas group, members may forage on resources separated by several hundred meters (Tutin 1996; Remis 1997; Doran & McNeillage 2001; Bermejo 2004). As previous primate studies suggest that group cohesion influences the use of long-range calls (Goodall 1986; Clark 1993; Harcourt et al. 1993; Mitani 1996), while close calls represent the bulk of mountain gorilla vocal output (Harcourt et al. 1993; Stewart & Harcourt 1994; Mitani 1996), western gorillas might also rely on long-range calls for within-group communication, due to reduced group cohesion.

Here, we present data on the full repertoire of wild western gorillas, with the goal of assessing which calls are limited to specific contexts and exploring the role they play in their vocal behavior. Therefore, the first aim of this study is to categorize and quantitatively describe their vocalizations. We then assess the behavioral contexts of western gorilla vocalizations

Table 1: Mountain gorilla vocal repertoire

Call type	Other names	Proposed contexts	M	F	J	I
Single grunt	Belch vocalization ²	WG; multiple ^{1,2,3}	x	x	x	x
Double grunt	Belch vocalization ²	WG; multiple ^{1,2,3}	x	x	x	x
Cough grunt	Staccato grunt ¹ ; pig grunt ² ; pant series	WG; aggression ^{1,2,3}	x	x		
Copulation grunt	Copulatory grunt ¹	WG; copulation ^{1,2,3}	x	x ^a (c)		
Grumble	Belch vocalization ²	WG; multiple ^{1,2,3}	x	x	x	x
Hum	Belch vocalization ²	WG; multiple ^{1,2,3}	x	x	x	x
Sing	High hum ²	WG; multiple ^{2,3}			x	x
Chuckle	Panting chuckle ¹	WG; play ^{1,2,3}				x
Whinny	Train grunt ³ ; neighing horse sound ^{1,2}	WG; multiple ^{1,3,4,5} ; idiosyncratic ²	x			
Hoot series	Ho ho ho ¹	EG; interaction with extra-group male gorillas or humans ^{1,2,3}	x			
Hoot-chest beat		EG; interaction with extra-group male gorillas or humans ^{1,2,3}	x	x ^a (1)		
Whimper	Whine ¹	WG; abandoned, injured, in danger ¹ ; distress ²		x ^a	x ^a	x ^a
Cry	Infant scream: tantrum ²	WG; left behind, in danger ¹ ; distress ²				x ^a
Scream		WG/EG; aggression ^{1,2} ; alarm ² ; human presence ^{1,2}	x ^a	x ^a	x ^a	x ^a
Bark	Hoot-, question-, hiccup-bark ²	EG; aggression ¹ ; alarm ^{1,2} ; human presence ^{1,2}	>x ^a	x ^a	x ^a	x ^a
Roar	Threat-, alarm-bark ²	EG; attack ^{1,2} ; human presence ^{1,2}	>x ^a	x ^a		

Call types, proposed contexts, and age/sex differences [silverback/adult male (M), adult female (F), juvenile (J), infant (I)] in call usage. Context is distinguished between within group (WG) and extra group (EG) and whether the call type is used in one or more (multiple) contexts.

References: ¹Schaller 1963; ²Fossey 1972; ³Harcourt et al. 1993; ⁴Watts 1991a; ⁵Sicotte 1994

^aUnhabituated animals; (c) only captive animals; (1) heard only once; >mostly this age/sex class.

and compare them to those of mountain gorillas. Finally, we assess the occurrence of context-specific calls within the repertoire and vocal behavior of wild western gorillas (i.e., proportion within the repertoire and call rates). We discuss our results in light of current knowledge about mountain gorilla vocal behavior, to determine whether differences in call production and/or usage exist between the two species and, if so, whether they are explained by differences in gorilla socioecology.

Methods

Subjects and Study Area

We collected behavioral data and vocal recordings over a 19-month period (June–Aug. 2007; Mar. 2009–June 2010) from a single, well-habituated group of wild western gorillas at the Mondika Research Center (02°21'859"N; 016°16'465"E), Central African Republic and Republic of Congo (Doran-Sheehy et al. 2004, 2007). Throughout most of the study, the group included the same adult silverback male, four adult females, and two older infants (4.5 and 5 yr old in 2009). Three individuals were present for briefer periods: one female was present in 2007 but not in 2009 (death), one female was present in 2009–10 but not in 2007 (immigration), and one 6- to 7-yr-old juvenile female was present in 2007 but not in 2009–10 (natal dispersal). From Nov. 2009

to May 2010, we also collected data for one silverback male, one adult female, and one infant of a second less habituated group.

Data Collection

Two teams of two researchers each (led by RS) collected simultaneous 2-h focal animal samples (Altmann 1974) of the habituated silverback and one of the group's four adult females or, less frequently, juveniles or infants. Attempts were made to rotate sampling evenly among the females and, because rates of vocalization may vary with time of day (e.g., Tenaza & Tilson 1977; Sekulic 1982; Egnor et al. 2007), to sample individuals equally across four time periods (7:00–9:30; 9:30–12:00; 12:00–14:30; 14:30–17:00). Total sampling of the habituated group included 533 focal follows, which were collected during 1572 contact hours (Table 2). This data set includes 170 male follows (764 h), 319 female follows (695 h), including an average of $80 \pm \text{SD } 6$ follows and $173.8 \pm \text{SD } 14.5$ h per female, 12 juvenile follows (54 h), and 32 infant follows (59 h; Table 2). The male in the less habituated group was observed for 54 h during 17 focal follows of 2–4 h each. However, we used only the recordings of his calls for the acoustic analysis and did not use the behavioral data collected, because previous studies demonstrated that habituation level influences vocal behavior (Fossey 1972; Harcourt et al. 1986, 1993).

Table 2: Data set for western gorilla silverbacks/adult males (M), adult females (F), juvenile (J), and infants (I)

Individual	N _{FOLLOW}	T _{SAMPLING} (hrs)	T _{CONTINUOUS} (hrs)	N _{VOCs}	N _{CALLS}	R _{TIME-PERIOD}
Habituated group						
M1	170	764	462	8344	1081	1.0:1.0:1.1:1.0
F1	84	176	87	1149	401	0.9:1.0:1.1:1.0
F2	86	189	93	627	154	1.1:1.0:0.9:1.0
F3	75	154	72	228	75	1.0:0.9:1.2:0.9
F4	74	176	82	310	36	1.0:1.0:1.1:1.0
F5	—	—	—	—	17	1.0:1.0:1.0:1.0
F6	—	—	—	—	42	—
J1	12	54	53	296	45	1.0:1.0:1.0:1.0
I1	18	32	23	300	154	0.9:0.9:1.4:0.7
I2	14	28	19	106	50	0.8:0.8:1.2:1.2
Subtotal	533	1572	890	11 360	2055	
Semi-habituated group						
M2	17	54	54	378	68	0.9:1.1:1.1:0.9
F7	—	—	—	—	29	—
I3	—	—	—	—	11	—
Total	550	1625	944	11 738	2163	

These include number of focal follows (N_{FOLLOW}), total focal sampling time (T_{SAMPLING} (hrs)), total continuous sampling time (T_{CONTINUOUS} (hrs)), number of vocalizations heard during focal follows (N_{VOCs}), number of vocalization recordings selected for acoustic analysis (N_{CALLS}), and ratio of time spent in each time period relative to the total follow time (R_{TIME-PERIOD}) [e.g., the juvenile (J1) was followed 12 times, three times in each of the four time period 3/3:3/3:3/3:3 = 1:1:1:1].

Each focal sampling hour was divided into six 5-min periods of continuous sampling (0–4; 10–14; 20–24; 30–34; 40–44; 50–54; Martin & Bateson 1993) alternated with six 5-min periods of scan sampling (data used for a separate study). During continuous sampling, we noted all focal animal vocalizations, coding each as one of 17 call types, distinguishable by ear and corresponding to the 16 described previously for mountain gorillas (Schaller 1963; Fossey 1972; Harcourt et al. 1993; Table 1) plus a call type (sex-whinny) not described in mountain gorillas. When the observer could not identify the call, it was recorded as a ‘miscellaneous’ call type. For each vocalization, we recorded one of 14 mutually exclusive contexts, which were modified after contexts identified previously for mountain gorillas (Table 1). These contexts included (1) within-group aggression; (2) alarm/alertness; (3) within-group display; (4) extra-group display; (5) mother–infant nursing interaction (when the focal individual was the infant), which was divided into two contexts (6) nipple contact and (7) nipple conflict; (8) foraging; (9) play; (10) resting; (11) pre-mating courtship behavior; (12) sex; (13) travel pause; (14) traveling; and (15) other behaviors (see Data S1, Supporting information for a detailed description of contexts). We also recorded focal animal activity at 1-min instantaneous sampling intervals, using the same ethogram to provide a baseline measure of activity. At the beginning of the study, we

computed interobserver reliability between the two team leaders in the identification of the 17 call types and 14 behavioral contexts by means of Cohen’s kappa (Lehner 1996). We included data in the analysis only after consistency between observers was confirmed ($\kappa \geq 0.90$).

During continuous focal animal sampling periods, one team (led by RS) made digital recordings of all focal vocalizations (at a distance of <10 m), using a portable Professional Solid State Recorder Marantz PMD671 and a Sennheiser MKH 416 short shotgun microphone, with a modular suspension (shock mount) and protected by a foam windshield (MZW415ANT). To increase the sample of rare vocalizations (i.e., hoot series, screams, and barks) used in the acoustic analysis, we also recorded 167 calls opportunistically, from females 5 and 6 in the habituated group, and from male 2, female 7, and infant 3 in the less habituated group (Table 2).

Acoustic Analysis

Digital recordings were made in mono at a sampling frequency of 48 kHz and saved as PCM format. We converted the sampling frequency of calls to one of three frequencies (22 050 Hz, 11 025 Hz, or 4000 Hz), depending on the call frequency range, using the software Avisoft SASLab Pro 5 (R. Specht, Berlin, Germany). We used a frequency resolution of 1024 points

for the fast Fourier transform analysis and worked with the Hamming window to generate spectrograms of each call (Avisoft SASLab Pro 5). Of the 11 738 calls recorded, we selected 2163 for further acoustic analysis based on the quality of spectrographs (i.e., low call/background noise ratio; Table 2).

Gorilla vocalizations are typically composed of one or more call segments, each of which may be used once or more than once. For example, a bark is composed of a single segment (a loud, harsh sound) that could be given once ('bark') or several times ('bark, bark, bark'). A copulation grunt is also composed of a single segment (a soft, guttural call), although it is always emitted in long sequences of more than 80 repeats ('huh, huh, huh...'). A double grunt is a soft vocalization that is composed of two distinct segments, which may be given once ('uhm-humh') or several times ('uhm-humh, uhm-humh, uhm-humh'). We considered the minimum number of unique call segments that characterize a particular vocalization as the unit for acoustic analysis regardless of how often they are repeated (following Bezerra et al. 2010). Thus, in the three examples described above, the unit of acoustic analysis would be a single bark (bark), a single copulation grunt (huh) and a single two-syllabled double grunt ('uhm-humh').

We used a custom software program (LMA), developed by Kurt Hammerschmidt (1990), to calculate a suite of acoustic parameters (Schrader & Hammersch-

midt 1997; Fischer et al. 2013), excluding those that have been shown to be highly influenced by external factors such as habitat structure and recording conditions (i.e., call start and end parameters: Maciej et al. 2011). We use the selected 28 parameters related to time, frequency, dominant frequency bands measurements, global modulation, relative amplitude characteristics, and tonality (see Data S2, Supporting information) in the statistical analyses of vocalizations.

Statistical Analysis—Call Classification

We conducted acoustic analysis on 14 of the 17 call types (Table 3), excluding three call types (chuckles, roars, and barks) because of limited ($n < 6$) sample size or quality of acoustic recordings. We tested whether the call types were acoustically distinct by performing a stepwise procedure discriminant function analysis (DFA; Klecka 1980; with F -value thresholds set for acceptance or rejection of independent variables at $F = 3.84$ and $F = 2.71$) and a leave-one-out cross-validation DFA (Fischer et al. 2001; Wich et al. 2003; Barros et al. 2011) of the 28 acoustic parameters. We present the results from both the stepwise DFA and cross-validated DFA, because the similarity of the results indicates that the profiles derived from calls are stable (Barros et al. 2011). We identified which acoustic parameters were significant in distinguishing

Table 3: Summary of western gorilla vocal repertoire. For each call type the following are indicated: (1) call distance category [close-range (C) or long-range (L)], (2) call specificity [signal and/or context specific (S) or general (G)], (3) context, if specific, and whether is used in within- (WG) or extra-group communication, and (4) call rate for male (M), female (F), juvenile (J), and infant (I) gorilla

Repertoire				Call rates			
Call	Distance	Specificity	Context	M	F	J	I
Single grunt	C	G	WG; multiple	3.62 ± 2.4	2.91 ± 1.4	0.22 ± 0.3	0.88 ± 0.6
Double grunt	C	G	WG; multiple	10.42 ± 1.1	0.71 ± 0.7	0.01 ± 0.0	0.08 ± 0.1
Cough grunt	C	S	WG; aggression	0.55 ± 0.2	0.29 ± 0.2	0.04 ± 0.1	0.14 ± 0.1
Copulation grunt	C	S	WG; copulation	0.56 ± 0.9	0.01 ± 0.1		
Grumble	C	G	WG; multiple	1.68 ± 1.0	1.01 ± 0.8	1.60 ± 1.5	2.41 ± 0.8
Hum	C	S	WG; foraging	0.60 ± 0.4	1.38 ± 1.2	2.00 ± 2.8	0.83 ± 0.9
Sing	C	S	WG; foraging			0.89 ± 0.8	
Chuckle	C	S	WG; playing	0 ^a	0.03 ± 0.1	0.63 ± 1.0	1.68 ± 4.1
Whinny	C	G	WG; multiple	0.60 ± 0.3			
Hoot series	L	G	WG; multiple	0.56 ± 0.4	0.08 ± 0.1	0.09 ± 0.3	0.51 ± 0.61
Hoot-chest beat	L	S	EG; interaction	0.05 ± 0.1			
Whimper	C	S	WG; nursing attempt				1.75 ± 0.1
Cry	L	S	WG; nursing denied				0.24 ± 0.1
Scream	L	S	WG; aggression; alarm		0.03 ± 0.1	0	
Bark	L	S	EG; alarm; human presence	0.16 ± 0.03			
Roar	L	S	EG; attack; human presence	0 ^b			
Sex-whinny	C	S	WG: mate- courtship	0.16 ± 0.3			

^aRare and only observed during non-focal sampling.

^bOnly heard from non-habituated animals.

call types, by running linear mixed models (LMMs; West et al. 2006) of each parameter that was significantly correlated ($r > 0.5$) to the functions generated by DFA. Within this analysis, we tested for differences across call types, while controlling for random effects (i.e., repeated and unequal sampling of individuals). To adjust for multiple analyses, we used the Hochberg correction (Hochberg 1988). We conducted all statistical analyses using SPSS 19.0 (IBM Corp., Armonk, NY, USA).

When calls were misclassified, we examined whether they were likely to be assigned to call types with similar acoustic structure. To do so, we coded each call type as one of nine broad acoustic categories (grunts, grumbles, singing, whinnies, whimpers, laughs, screams, barks, and hoots) on the basis of shared similarities in the spectrograms, following Beeman's (1998) acoustic classification of animal vocalizations, which were modified from the four principal acoustic groups identified in apes (i.e., grunts, screams, barks, and hoots). The number of call types within a broad acoustic category ranged from 1 to 4 (see Data S3, Supporting information for detailed description of both call types and acoustic categories).

Behavioral Data Analysis

From the focal data, we assigned 99.09% of calls ($n = 11\,053$) to one of 17 call types that we could identify by ear and spectrographic analysis. We could not classify the remaining 0.9% of calls, because they were not heard clearly or were included in sequences of multiple call types (i.e., miscellaneous call type). Although the latter is a potentially intriguing area of study, we did not consider it further.

As a first step toward assessing the potential function of call types, we examined the context in which the 11 053 focal calls were used, following Seyfarth et al. (1980). We classified a call type as context specific if it was given in the same behavioral context more than 75% of the time. Because multiple call types can be used in the same context, we then examined whether this was the primary call type for this context, classifying it as signal specific if it accounted for more than 75% of all calls given during that context (Seyfarth et al. 1980). We classified call types that were signal and/or context specific as 'specific calls' and call types that were neither context nor signal specific as 'general calls'. We determined the daily rate of each call type for each individual from focal follows, averaging the daily rates across months, before averaging monthly averages across the 12 mo to obtain the overall rates. We used a one-way ANO-

VA (Miller 1997) to test for individual differences in adult calling rates (one male and four females). To assess the relative contribution of general vs. specific calls, we coded each call type as either general or specific and then summed the rate of each call type within each category.

Results

Vocal Repertoire Based on Acoustic Characteristics and Usage

In the field, we distinguished 17 call types based on acoustic characteristics and their contextual use (Table 3 and Fig. 1). These include the 16 call types reported previously for mountain gorillas (Table 1) and one additional call type, the sex-whinny. We named call types following the nomenclature used for mountain gorillas (Table 3; further description is provided in Data S3, Supporting information). The 17 call types included both close-range (i.e., soft, low amplitude) and long-range (i.e., loud) calls, the latter used for within- and extra-group communication (Table 3). Close-range calls included single, double, cough, copulation grunts (Table 3; Fig. 1a), grumbles, hums (Fig. 1b), singing (Fig. 1c), two types of horse-like sounds or whinnies (Fig. 1d), whimpers (Fig. 1f), and chuckles. Long-range calls were comprised of screams, infant cries (Fig. 1g), barks, roars, and two hooting calls, with sequences of $15 \pm \text{SD } 7$ ($n = 89$; range = 3–44) tonal segments, distinguished by whether or not they were followed by chest beats (Fig. 1e1 and 2).

Age-sex classes seem to differ in the number of calls in their vocal repertoire (Table 3). As in mountain gorillas, the silverback male had the widest repertoire, using 12 call types, including five used exclusively by him (whinny, sex-whinny, hoot-chest beat, roar, and bark; Table 3). Adult females used nine call types, including one call (scream) used uniquely by adult and juvenile females (Table 3). Immature gorillas, including both infants and juveniles, used nine call types (Table 3), three of which (sing, cry and whimper) were not used by adults.

Structural Examination of Vocal Repertoire (DFA)

Discriminant function analysis distinguished among the 14 call types tested. Correct assignments were made more often than expected by chance (7.14%), with a classification accuracy of 63% and 62% for DFA and cross-validated DFA analyses, respectively (*Wilks'* $\lambda = 0.001$; $\chi^2 = 14,149.1$, $p < 0.001$). DFA generated

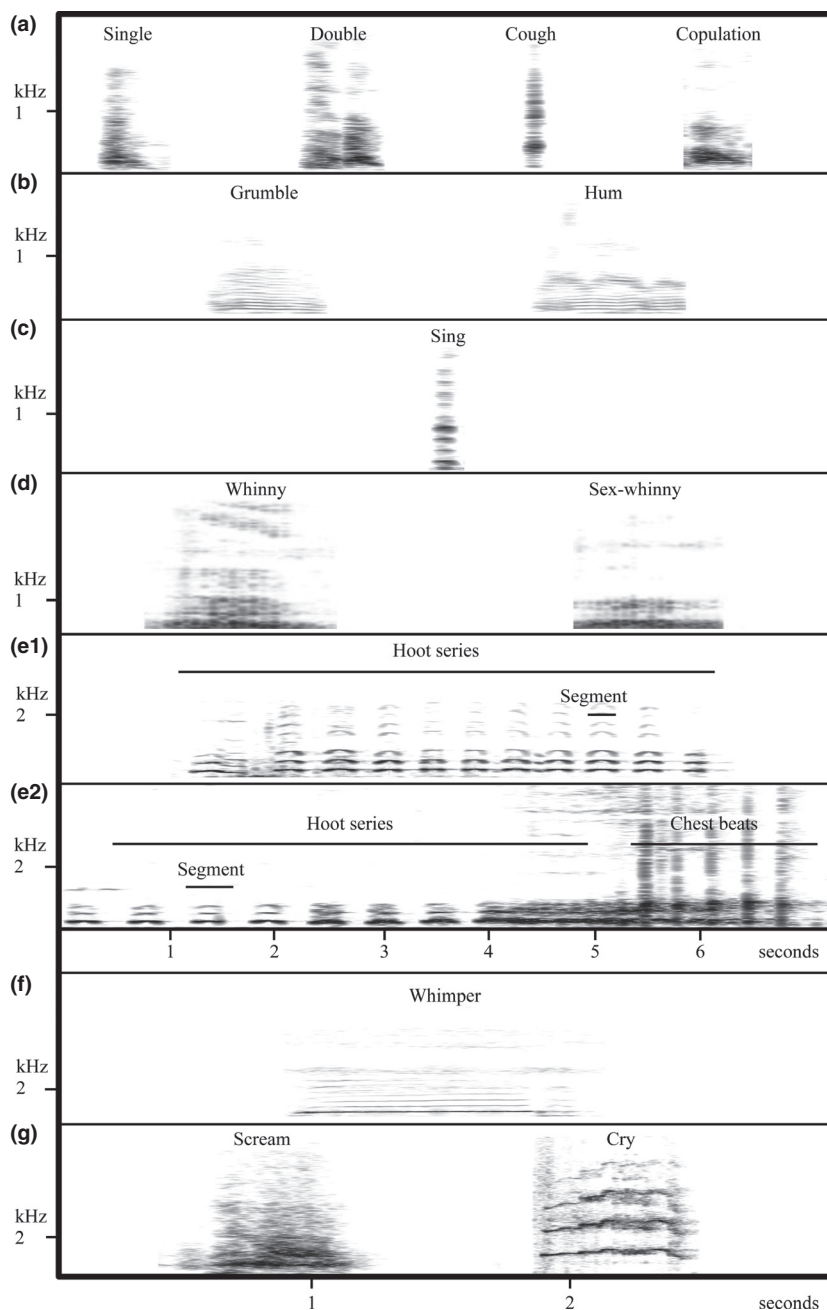


Fig. 1: Spectrograms of 14 western gorilla call types, grouped in seven broad acoustic categories, including the following: grunts (a), grumbles (b), singing (c), whinnies (d), hoots (e1 and 2), whimper (f), and screams (g).

seven canonical discriminate functions that explained more than 2% of the variance. The first three functions explained 84% of variance and were the only functions that had eigenvalues greater than one (eigenvalues: $F1 = 8.1$; $F2 = 3.2$; $F3 = 1.6$). Function 1 was primarily correlated with frequencies and pitch parameters (i.e., $df1_{mean}$, $dfa1_{mean}$, $dfa2_{mean}$, $fp1_{mean}$, and pf_{mean} ; description of parameters in Data S2, Supporting information) and accounted for 53% of the variance. Function 2 was correlated with

call duration and accounted for 21% of the variance. Function 3 was correlated with the maximum value of the first frequency in the call with more energy ($df1_{max}$) and explained 10% of variance. The seven acoustic parameters used to generate the DFA differed across call types when tested using LMMs and after controlling for multiple analyses. The percentage of calls assigned correctly to each call type (Table 4) by cross-validated DFA was 66% on average, ranging from 39% for copulation grunts to 89% for singing

(DFA: 40–89%). All calls, except copulation grunts, grumbles, and hums, achieved $\geq 55\%$ correct classification (Table 4). Mean values of most significant acoustic parameters for all call types are shown in Table 5.

Misclassified calls were almost always assigned within the same broad acoustic category (Table 4; Fig. 2). For example, 38% of single grunts and 61% of copulation grunts were misclassified, but most of these calls (single grunt: 20%; copulation grunt: 48%) were classified as other grunts (Table 4).

Context- and Signal-Specificity

Twelve of 17 call types (71%) were context specific (i.e., given mostly in one specific context), including cough grunts (during aggression), copulation grunts (during mating), humming and singing (during foraging), sex-whinnies (during male courtship), infant whimpers (attempting nipple contact), infant cries (when nipple contact was rejected), screams (during escalated aggression), barks (when alarmed), chuckles (during play), and the hoot-chest beats (during within- and between-group displays; Table 6). Additionally, roars were considered context specific because they always occurred in a single context (when the silverback charged humans during the habituation process). However, because they were only heard during *ad libitum* sampling of unhabituated gorillas, this call is not included in Table 6. All

context-specific calls were given in a single context more than 97% of the time, except for hums, which were given in a context-specific way 91% of the time (Table 6).

Eight context-specific call types were also signal specific (i.e., the most common call type used in a specific context; Table 7), including cough grunts, copulation grunts, sex-whinnies, hoot-chest beats, whimpers, cries, screams, and play chuckles. The other context-specific call types (barks/roars, humming, and singing) were given in a single context (alarm in the former and feeding in the latter), but were not the only call types given during a specific activity. Thus, a total of eight call types (cough grunts, copulation grunts, sex-whinnies, hoot-chest beats, whimpers, cries, screams, and chuckles) showed both signal and context-specificity. Five call types (single grunt, double grunt, grumble, and non-sex-whinnies) were neither context-specific (Table 6) nor signal specific (Table 7) and were considered 'general calls' (Table 3).

Most call types used exclusively by a single age or sex class (87.5%) were context and/or signal specific. Context-specific call types used uniquely by the male appear to function in group defense (hoot-chest beat, bark, and roar) and courtship (sex-whinny). Context-specific calls used only by females occurred during escalated within-group aggression (scream) and by infants in response to distress related to weaning (whimper, cry).

Table 4: Percent of calls assigned to each of 14 call types by cross-validated DFA

Call type	Cross-validated DFA: Predicted group membership													
	SG	DG	CG	COG	GR	HM	SI	WP	WH	SWH	HCB	HO	SC	CR
SG	62	17	2	1	4	4	0	0	0	0	0	0	0	0
DG	24	65	3	5	9	1	2	0	0	0	1	0	0	0
CG	11	0	66	14	0	0	8	0	0	0	1	0	0	0
COG	18	18	12	39	0	0	8	0	0	0	5	1	0	0
GR	20	1	0	0	40	38	1	0	0	0	0	0	0	0
HM	9	3	1	1	39	44	2	0	0	0	0	0	0	0
SI	0	4	0	4	4	0	89	0	0	0	0	0	0	0
WP	0	0	0	0	0	0	5	59	0	2	0	32	0	2
WH	0	3	0	0	3	10	0	3	70	7	3	0	0	0
SWH	0	0	0	0	0	0	0	18	27	55	0	0	0	0
HCB	1	0	1	3	0	0	6	1	0	0	79	8	0	0
HO	0	1	0	3	0	0	3	0	0	0	34	58	0	0
SC	0	0	4	0	0	0	0	0	8	0	4	0	60	24
CR	0	0	0	0	0	0	0	9	0	0	2	0	8	81

DFA, discriminant function analysis.

Call type abbreviation include SG, single grunt; DG, double grunt; CG, cough grunt; COG, copulation grunt; GR, grumble; HM, hum; SI, sing; WP, whimper; WH, whinny; SWH, sex-whinny; HCB, hoot-chest beats; HO, hoot series; SC, scream; CR, cry. Bolded numbers indicate the percentage of calls assigned correctly to each call type. Call types are grouped in call categories (i.e., call types with similar broad acoustic structure), which are divided by solid lines.

Table 5: Mean ($\pm$ SE) of the major acoustic parameters that distinguished among the 14 call types in adult male (M) and female (F) western gorillas as well as those of juvenile (J) and infant (I) for the few calls uniquely given by immatures. The number of call segments used in the analysis is indicated by N_c

Call	A-S	N_c	Duration	Tonality	dfa2mean	df1mean	df2mean	fp1mean	fp1amean	pymax
Double grunt	M	171	838.3 $\pm$ 25	1.8 $\pm$ 0.3	227.7 $\pm$ 5	109.9 $\pm$ 1	165.7 $\pm$ 2	151.3 $\pm$ 3	759.7 $\pm$ 19	244.5 $\pm$ 8
grunt	F	56	668.4 $\pm$ 52	4.0 $\pm$ 1	304.5 $\pm$ 14	94.5 $\pm$ 2	166.6 $\pm$ 5	114.0 $\pm$ 6	221.2 $\pm$ 16	273.0 $\pm$ 26
Single grunt	M	74	785.0 $\pm$ 56	5.8 $\pm$ 1	197.8 $\pm$ 6	112.6 $\pm$ 2	182.3 $\pm$ 6	121.9 $\pm$ 3	581.3 $\pm$ 30	186.2 $\pm$ 7
	F	76	495.4 $\pm$ 35	4.1 $\pm$ 1	316.5 $\pm$ 13	91.0 $\pm$ 1	174.1 $\pm$ 7	102.8 $\pm$ 5	240.3 $\pm$ 15	225.7 $\pm$ 26
Cough grunt	M	70	209.5 $\pm$ 1.5	16.8 $\pm$ 4	357.9 $\pm$ 16	99.4 $\pm$ 2	169.1 $\pm$ 6	144.8 $\pm$ 10	263.2 $\pm$ 16	246.4 $\pm$ 19
	F	156	206.5 $\pm$ 0.9	11.4 $\pm$ 2	407.6 $\pm$ 16	98.7 $\pm$ 2	174.0 $\pm$ 4	188.1 $\pm$ 10	240.4 $\pm$ 10	300.4 $\pm$ 19
Cop. grunt	M	73	416.1 $\pm$ 28	5.4 $\pm$ 2	236.5 $\pm$ 5	113.8 $\pm$ 3	170.3 $\pm$ 4	173.0 $\pm$ 6	615.9 $\pm$ 33	218.6 $\pm$ 10
	F	73	224.5 $\pm$ 6	20.7 $\pm$ 4	288.5 $\pm$ 4	129.3 $\pm$ 7	242.8 $\pm$ 23	232.0 $\pm$ 9	396.9 $\pm$ 231	271.6 $\pm$ 9
Grumble	M	46	1568.1 $\pm$ 74	22.1 $\pm$ 4	208.5 $\pm$ 7	99.8 $\pm$ 2	192.8 $\pm$ 7.4	78.56 $\pm$ 4	205.5 $\pm$ 13	175.7 $\pm$ 14
	F	34	2068.8 $\pm$ 206	35.7 $\pm$ 4	274.7 $\pm$ 11	88.5 $\pm$ 1	150.9 $\pm$ 4.7	83.7 $\pm$ 7	164.7 $\pm$ 20	266.6 $\pm$ 27
Hum	M	45	1817.2 $\pm$ 88	8.9 $\pm$ 2	243.8 $\pm$ 7	96.9 $\pm$ 2	158.5 $\pm$ 3.8	93.1 $\pm$ 4	239.0 $\pm$ 27	249.1 $\pm$ 16
	F	44	1971.2 $\pm$ 170	42.3 $\pm$ 5	292.2 $\pm$ 16	87.1 $\pm$ 3	144.0 $\pm$ 7.1	83.3 $\pm$ 9	209.7 $\pm$ 16	224.7 $\pm$ 29
Sing	J	26	373.9 $\pm$ 52	70.0 $\pm$ 4	327.5 $\pm$ 25	120.6 $\pm$ 6	249.7 $\pm$ 11	129.9 $\pm$ 11	139.0 $\pm$ 14	213.0 $\pm$ 30
Whimper	I	44	687.8 $\pm$ 70	75.6 $\pm$ 3	1357.3 $\pm$ 106	460.7 $\pm$ 13	1341.2 $\pm$ 90	429.8 $\pm$ 24	300.6 $\pm$ 28	2748.9 $\pm$ 35
Cry	I	62	609.5 $\pm$ 39	57.1 $\pm$ 4	2649.8 $\pm$ 73	1235.7 $\pm$ 35	2302.6 $\pm$ 72	1477.9 $\pm$ 50	422.58 $\pm$ 28	3840.7 $\pm$ 24
Scream	F	25	911.6 $\pm$ 288	11.9 $\pm$ 7	1613.6 $\pm$ 124	616.6 $\pm$ 61	1044.5 $\pm$ 71	1048.1 $\pm$ 10	724.2 $\pm$ 111	1608.9 $\pm$ 11
Whinny	M	30	1848.6 $\pm$ 92	21.0 $\pm$ 3	387.0 $\pm$ 64	196.2 $\pm$ 8	573.39 $\pm$ 34	73.35 $\pm$ 5	898.8 $\pm$ 60	753.8 $\pm$ 23
Sex-whinny	M	11	1680.3 $\pm$ 208	7.4 $\pm$ 3	1417.3 $\pm$ 27	330.4 $\pm$ 34	1010.8 $\pm$ 97	111.4 $\pm$ 24	342.6 $\pm$ 77	2799.3 $\pm$ 50
Hoot-cb	M	316	158.6 $\pm$ 3	64.2 $\pm$ 2	308.4 $\pm$ 11	208.3 $\pm$ 1	601.5 $\pm$ 31	154.2 $\pm$ 4	659.0 $\pm$ 21	244.4 $\pm$ 9
Hoot series	M	403	171.1 $\pm$ 3	63.7 $\pm$ 2	282.9 $\pm$ 5	214.1 $\pm$ 2	529.0 $\pm$ 26	170.6 $\pm$ 4	734.8 $\pm$ 19	247.2 $\pm$ 4
	F	328	222.2 $\pm$ 12	58.1 $\pm$ 2	478.2 $\pm$ 9	347.8 $\pm$ 3	749.6 $\pm$ 24	366.4 $\pm$ 4	606.4 $\pm$ 15	414.0 $\pm$ 6

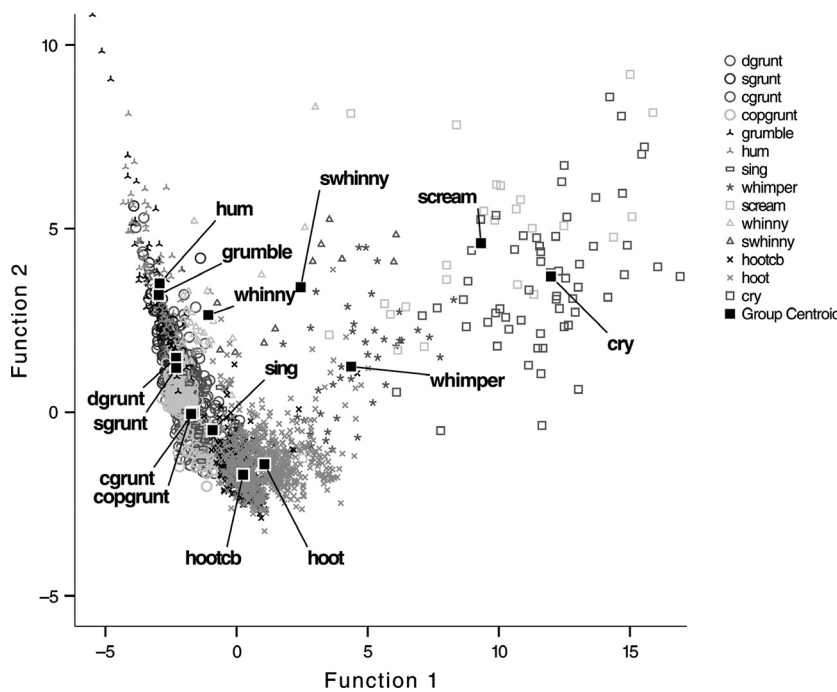


Fig. 2: Distribution of the discriminant scores for the 14 call types (abbreviated call types: sgrunt = single grunt; dgrunt = double grunt; cgrunt = cough grunt; copgrunt = copulation grunt; swinny = sex-whinny; hoot = hoot series; hootcb = hoot-chest beat).

Occurrence of Specific and General Calls

On average, adult western gorillas vocalized at a rate of $8.9 \pm \text{SD } 6.3$ calls/h ($n = 5$ individuals, range 18.7–3.0 calls/h), although call rates varied among individu-

als ($\text{ANOVA}_{\text{adult}} F = 92.27$; $df = 4$; $p < 0.001$). The mean overall calling rate of the male ($18.6 \pm \text{SD } 3.1$ calls/h) was roughly triple that of the mean rate of females ($6.5 \pm \text{SD } 4.2$ calls/h) or immatures ($7.3 \pm \text{SD } 4.3$ calls/h; Fig. 3). The calling rate for

Table 6: Call context-specificity: the percent of calls given in each of context Contexts include aggression (AG), alarm (AL), display between group (DPE) and within group (DP-WG), forage (FO), courtship (CO), sex (SE), play (PL), rest (RE), travel pose (TRP), travel (TR), seek nipple contact (NPA), and nipple contact rejected (NPR). Context-specific calls (i.e., for which $\geq 75\%$) are indicated in bold. Sample size (N_{VOC}) includes total number of each call type heard during continuous focal follows

Calls/sounds		Contexts												
Calls	N _{Voc}	AG	AL	DPE	DPW	FO	CO	SE	PL	RE	TRP	TR	NPA	NPR
Single grunt	2776	0	0	0	0	48	0	0	0	29	11	12	0	0
Double grunt	4944	0	0	0	0	59	0	0	0	24	8	9	0	0
Cough grunt	361	100	0	0	0	0	0	0	0	0	0	0	0	0
Cop. grunt	29	0	0	0	0	0	0	100	0	0	0	0	0	0
Grumble	1192	0	0	0	0	26	0	0	0	22	23	28	0	1
Hum	836	0	0	0	0	91	0	0	0	3	1	5	0	0
Sing	26	0	0	0	0	100	0	0	0	0	0	0	0	0
Sex-whinny	64	0	0	0	0	0	97	0	0	0	3	0	0	0
Whinny	302	0	1	0	1	6	2	0	0	37	47	6	0	0
Hoot-chest beats	14	0	0	64	36	0	0	0	0	0	0	0	0	0
Hoot series	256	0	3	0	3	23	0	0	0	49	18	4	0	0
Whimper	83	0	0	0	0	0	0	0	0	0	0	0	100	0
Cry	12	0	0	0	0	0	0	0	0	0	0	0	1	99
Scream	1 ^a	100	0	0	0	0	0	0	0	0	0	0	0	0
Bark	10	0	100	0	0	0	0	0	0	0	0	0	0	0
Chuckle	35	0	0	0	0	0	0	0	100	0	0	0	0	0
Miscellaneous	112	0	0	0	0	38	1	0	0	38	12	11	0	0

^aAll other females also uttered screams during aggression, however, during non-focal observation. Part of those non-focal observations were recorded and used in the acoustic analysis.

Table 7: Call signal-specificity: the percent of 14 call types given in each context

Context	N_{CC}	N_{HR}	SG	DG	CG	CPG	GR	HM	SNG	SWH	WH	HC	HS	WP	CRY	SC	BK	CK	MI
Aggression	362	5.9	0	0	100	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Alarm	27	2	7	7	0	0	7	0	0	0	11	0	30	0	0	0	38	0	0
Display extra group	9	0.6	0	0	0	0	0	0	0	0	0	100	0	0	0	0	0	0	0
Display within group	17	0.7	0	6	0	0	0	0	0	0	24	29	41	0	0	0	0	0	0
Forage	5441	435.4	24	54	0	0	6	14	0	0	0	0	1	0	0	0	0	0	1
Courtship	75	2	5	3	0	1	0	0	0	83	7	0	0	0	0	0	0	0	1
Sex	29	0.5	0	0	0	100	0	0	0	0	0	0	0	0	0	0	0	0	0
Play	41	1.7	7	0	0	0	7	0	0	0	0	0	0	0	0	0	0	86	0
Rest	2583	262.8	31	47	0	0	10	1	0	0	4	0	5	0	0	0	0	0	2
Travel pose	1149	44.9	26	32	0	0	24	1	0	0	12	0	4	0	0	0	0	0	1
Travel	1214	75.3	28	37	0	0	28	4	0	0	1	0	1	0	0	0	0	0	1
Nipple contact	93	2.3	3	0	0	0	8	1	0	0	0	0	0	88	0	0	0	0	0
Nipple conflict	13	0.3	0	0	0	0	0	0	0	0	0	0	0	8	92	0	0	0	0
N_{VOC}	11053	834.4	2776	4944	361	29	1192	836	26	64	302	14	256	83	12	1 ^a	10	35	112

Vocalizations include SG, single grunt; DG, double grunt; CG, cough grunt; CPG, copulation grunt; GR, grumble; HM, hum; SNG, sing; WH, whinny; SWH, sex-whinny; HC, hoot-chest beats; HS, hoot series; WP, whimper; CRY, cry; SC, scream; BK, bark; CK, chuckle; MI, miscellaneous calls. Signal-specific calls (i.e., those for which $\geq 75\%$ are given in a single context) are indicated in bold. Sample size is based on the number of calls given in each behavioral context (N_{CC}). The amount of time spent in each context (N_{HR}) and the total number of each call type (N_{VOC}) are also noted.

^aAll other females also uttered screams during aggression, however, during non-focal observation. Part of those non-focal observations were recorded and used in the acoustic analysis.

specific calls (i.e., the sum of call rates of all context- and/or signal-specific call types) was similar between the male (2.5 calls/h) and female ($1.8 \pm \text{SD } 1.4$ calls/h), whereas immatures had the highest specific calling rate ($4.1 \pm \text{SD } 0.8$ calls/h). For the 6-yr-old juvenile,

this rate resulted from relatively frequent humming, singing, and grumbling, whereas for infants, the common call types were primarily related to play and weaning (i.e., weaning/nipple contact; Table 3). The male's calling rate for general calls (16.1 calls/h) was

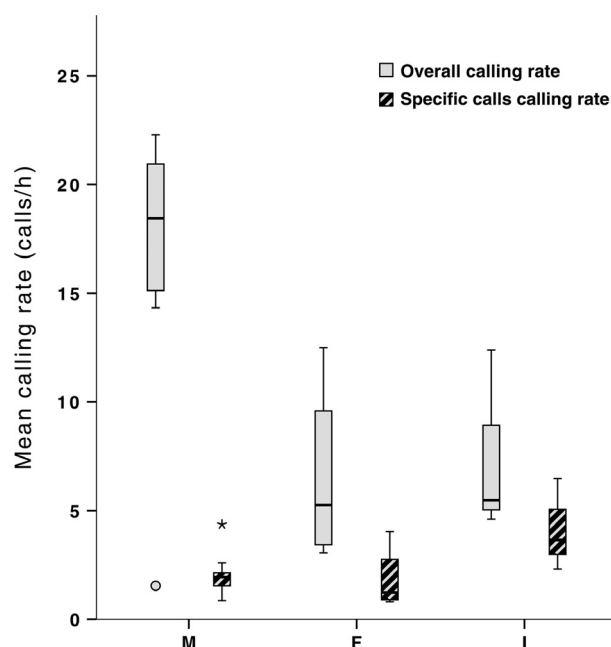


Fig. 3: Overall call rate and sum rate of specific call types. Male (M), female (F), immature [including both juvenile (J) and infant (I)] data were calculated from the mean monthly rates ($n_{M,F1-4} = 12$; $n_J = 2$, $n_{I1} = 10$, $n_{I2} = 7$), and monthly rates have been calculated by averaging across daily rates. Circle and star indicate outliers.

much higher than either that of the females ($4.7 \pm \text{SD } 2.8$ calls/h) or immatures ($2.67 \pm \text{SD } 1.1$ calls/h) and therefore accounted for the difference in overall calling rate. The male's higher general call rate was primarily the result of his more frequent use of double grunts, which were given 10.42 calls/h, and therefore more than 10 times the rate of other age-sex classes (Table 3).

Although specific call types (signal specific and/or context specific) account for a large part of the vocal repertoire (12 of 17; 71%), they contribute relatively little to the overall daily vocal production (Table 3). The degree to which this is true varies across age-sex classes (Fig. 3), with the proportion of specific calls contributing to the overall call production lowest in the male (13.5%), intermediate in females (26.8%), and highest in immatures (58.3%).

Discussion

Western Gorilla Vocal Repertoire

Western and mountain gorillas diverged recently (1.75 million years ago) with evidence of even more recent gene flow (Thalmann et al. 2007; Scally et al. 2012), which together might lead to an expectation of

high similarity between the two species. Accordingly, we found that the vocal behavior of the two species was very similar. We distinguished 17 call types produced by wild western gorillas on the basis of acoustic characteristics and behavioral contexts in which they were used. In both species, the silverback male has the widest repertoire and vocalizes more often than either females or immature individuals (mountain gorillas: Fossey 1972; Harcourt et al. 1993; western gorillas: this study), and each age-sex class uses some call types exclusively. As in mountain gorillas (Schaller 1963; Fossey 1972), the repertoire is fairly graded, with variation across call types. However, our analysis showed that call types were misclassified most exclusively within call types sharing similar acoustic properties (i.e., grunt-like sounds within themselves, horse-like sounds within themselves, grumbles with hums, etc.).

In some cases, we found call variation to be consistent with a general tendency of having specific acoustic characteristics linked to particular emotional states or levels of arousal experienced by the animal. Termed 'motivational structural' rules (MS; Morton 1977), this relationship between vocal communication and the caller emotional state has been documented in birds and many other mammals (Morton 1977; Owren & Rendall 2001), but remains controversial in primates (Hauser 1993). As predicted by the MS rules, low frequency, abrupt atonal western gorilla calls were produced when the caller was being aggressive both at close and long distances (i.e., cough grunts, barks), whereas moderate and high frequency tonal calls (i.e., singing, hoots, whimpers and screams) were produced in affiliative or stressful situations (i.e., maintenance activities and when distressed or being subject to aggression). In contrast, the structure of some gorilla calls and the motivational state in which they were given did not support the MS rules. Affiliative sounds were not always displayed with high frequency and pure tonal structure, also consistent with findings in other species (e.g., Peters 1984). For instance, western gorilla single and double grunts were produced during non-aggressive contexts (i.e., foraging, resting traveling) but were noisy calls of low frequencies. Furthermore, as suggested by Fichtel et al. (2001), calls produced in more aggressive circumstances, although exhibiting more non-harmonic energy (more noise in the call), displayed higher frequencies and peak frequencies (i.e., whimpers vs. cries; cough grunts vs. screams and barks).

Compared to mountain gorillas, western gorillas exhibit several noteworthy differences in their repertoire and usage of calls. For instance, their vocal

repertoire includes an additional call type, the sex-whinny, directed toward a specific receptive female, seemingly as an invitation to mate. Second, only females use screams, whereas, likely due to the presence of multiple males in their groups, both sexes in mountain gorillas give screams during within-group escalated aggression (Schaller 1963; Fossey 1972). Third, although both species produce the hoot series (long call), all western gorilla age-group classes use it during within-group communication, while just male mountain gorillas use it and only during intergroup encounters (Schaller 1963; Fossey 1972). Finally, western gorillas use a higher number of calls associated with specific contexts (i.e., humming, singing, and sex-whinny) than mountain gorillas (Schaller 1963; Fossey 1972; Harcourt et al. 1993), potentially explained by the limited visibility in their environment. Therefore, our study adds and complements to the data previously collected on mountain gorillas, building a detailed understanding of gorilla vocal communication, and how it varies between the two gorilla species.

Behavioral Context of Western Gorilla Vocalizations

A strong association between a vocal signal and a specific context provides the opportunity for listening individuals to gather information about the activity or context of unseen calling individuals and their surroundings. The use of distinct call types during specific contexts is extremely valuable, particularly for social species living in forested habitats (Morton 1975), allowing the coordination of group movements and activity patterns between individuals that are not in visual contact (e.g., Boinski 1991, 1993). Accordingly, more than half of the call types used by western gorillas were context and/or signal specific, with seven of these being given in a single clearly defined context that was similar to those described in mountain gorillas.

Compared to the mountain gorilla repertoire, a further three call types were found to be context and/or signal specific: humming, singing, and the sex-whinny. Humming and singing were interpreted in mountain gorillas as signs of contentment during a range of activities (i.e., feeding, resting, and traveling; Schaller 1963; Fossey 1972; Harcourt et al. 1993), rather than in any particular context. In contrast, the same call types were much more context specific in western gorillas, with individuals using them nearly exclusively during foraging. The third call, the sex-whinny, has not been documented for mountain gorillas, although a horse-like sound was

anecdotally described to occur in some cases before copulation (Watts 1991a). In contrast, in western gorillas, it was clearly identified as a separate call, based on its acoustic properties and the behavioral context in which it was primarily given (i.e., male courtship before copulation). As our study was mostly based on one adult male, this result should be confirmed by future works in other wild western gorilla groups, to exclude the possibility of the call being idiosyncratic to that specific male. If we assume that this call is part of the male western gorilla repertoire, the fact that it was not described in mountain gorillas raises interesting questions. Signaling an imminent copulation has different costs and benefits depending on the social organization of a species, as well as the age, sex, and rank of individuals under consideration (Pradhan et al. 2006). Although several hypotheses have been proposed to explain the function of pre-, during- and post-copulation calls, especially in female primates, their specific adaptive function and how this varies across species remains obscure (reviewed in Pradhan et al. 2006). Due to the long-shared evolutionary history and morphological constraints inherent in vocal production (Hammerschmidt & Fischer 2008), the sex-whinny might also be present in the mountain gorilla repertoire but the call might not have been described in their repertoire because it is rarely or never given or because its production was influenced by the level of the habituation (i.e., Schaller 1963; Fossey 1972). Alternatively, mountain gorillas might not use the call to avoid increasing competition between reproductively active males in the group (Bradley et al. 2005), a situation that does not occur in western gorilla bisexual groups, which have only one adult male (Gatti et al. 2004).

Only one call type, the hoot series, was more context specific in mountain than in western gorillas, given exclusively during male extra-group displays (Fossey 1972). However, in western gorillas, this call is produced during multiple non-aggressive contexts (e.g., traveling, foraging, and resting). The hoot series is exchanged within the group (R. Salmi, unpubl. data), and by all age-sex classes, suggesting that it might serve to maintain contact and regain proximity among dispersed group members. This condition is not experienced by mountain gorillas, in which group members are at all times in close proximity to each other (Watts 1991b). Further analysis of interindividual distances and its dynamics in relation to calls is needed to determine the role of the hoot series in maintaining western gorilla group cohesion.

Importance of Specific Signaling

Recent work on African ape vocal communication has focused on specific calls, providing deeper insights into their functions and potential referential meaning (e.g., Seyfarth et al. 1994; Stewart & Harcourt 1994; Crockford & Boesch 2003; Slocombe & Zuberbühler 2006; Slocombe et al. 2009; Clay & Zuberbühler 2011). As the majority of call types produced by western gorillas were context and/or signal specific, which are both prerequisites for referential signaling, future studies on these call types might reveal that also gorillas use referential communication as recently shown in other ape species (e.g., Slocombe et al. 2009; Clay & Zuberbühler 2011). As in mountain gorillas, the male is the center of the group, protecting both females and offspring from conspecific and non-conspecific danger (R. Salmi, pers. obs.; Watts 2000). Not surprisingly, calls uniquely used by one age–sex class were those given by the male during group defense or mating and by female and immature gorillas during alarm and distress situations.

Nevertheless, specific calls constituted <13–27% of total adult daily call production. Furthermore, while males used a larger number of specific call types, immature gorillas used them more frequently. Immature gorillas might benefit the most from the ability to precisely communicate distress to other group members, because they are the most vulnerable individuals in gorilla groups, with the highest mortality rate compared to all other age–sex classes (D. Doran-Sheehy, pers. obs.). Alternatively, because non-specific signals are given during many contexts, with their meaning potentially varying in relation to other factors, immature gorillas might require longer for social learning to be acquired and used correctly, as has been documented for other behaviors in apes (e.g., feeding behavior: Boesch & Boesch-Achermann 2000; Jaeggi et al. 2010; Byrne et al. 2011; cultural behavior: Matsuzawa et al. 2001; Lonsdorf et al. 2004). More information about the ontogeny of vocal production and its perception might help explain the observed differences in call usage across western gorilla age/sex classes.

General call types (i.e., grunts and grumbles), given during many contexts, were used most frequently, contributing more than 80% to the adult calling rate. They also presented more intermediate variants than other call types (R. Salmi, pers. obs.), raising the possibility that they convey different kinds of information (e.g., arousal and identity of the caller) and may include acoustically different subtypes, as described in other non-human primate species (e.g., Green 1975;

Seyfarth & Cheney 1984; Owren et al. 1997). Furthermore, studies on these general calls might facilitate the study of behavioral flexibility in both production and perception, which are considered crucial in gestural studies, but have received limited attention in primate vocal studies (Slocombe et al. 2011; Wheeler & Fischer 2012). The evolution of auditory genes has been linked to the evolution of language in humans (Clark et al. 2003), with the recent analysis of the gorilla genome indicating that several genes associated with hearing exhibit patterns of accelerated evolution similar to those in humans (Scally et al. 2012). Therefore, the study of gorilla vocal production and perception gains crucial importance in the interdisciplinary effort to ascertain the origin and evolution of human language.

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Supporting Information

Additional supporting information may be found in the online version of this article:

Data S1: Definition, code and description of behavioral contexts of wild western gorillas.

Data S2: Acoustic parameters of western gorillas included in the analysis.

Data S3: Description of western gorilla broad acoustic categories and call types.